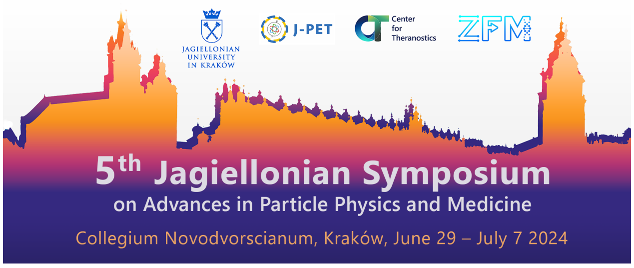

Abstract ID : **474**

Design, Synthesis, and Evaluation of Novel Gold Nanorod-Based Theranostic Agents for Anticancer Therapy

Content

Introduction

Pancreatic ductal adenocarcinoma (PDAC) is recognized as one of the most hypoxic and treatment-resistant tumor types. To address this challenge, we have designed AuNRs-GEM, a novel theranostic agent targeting PDAC. This agent combines the chemotherapeutic efficacy of gemcitabine with thermal therapy induced by near-infrared (NIR) light absorption.

The dual therapeutic approach leverages the benefits of both chemotherapy and localized hyperthermia. Gemcitabine, a standard chemotherapeutic agent, is known for its effectiveness against PDAC, while hyperthermia, induced by NIR absorption by gold nanorods (AuNRs), enhances drug delivery by increasing vessel dilation and altering tumor oxygen partial pressure (pO2). This multimodal treatment strategy aims to overcome the inherent resistance of hypoxic tumors to conventional therapies.

Additionally, AuNRs-GEM enables high-resolution imaging through photoacoustic imaging and computed tomography (CT), providing valuable insights into tumor morphology and treatment efficacy. By integrating these diagnostic and therapeutic modalities, we aim to improve the overall treatment outcomes for patients with PDAC.

Methods

The AuNRs-GEM nanohybrid was synthesized via amide bond formation between carboxylated gold nanorods and gemcitabine using the water-soluble carbodiimide method (EDCI-NHS). This nanohybrid was characterized using UV-VIS, FT-IR, and XPS spectroscopies. The concentration of AuNRs-GEM was established using RP-HPLC and TXRF techniques. The morphology and structure of the gold nanomaterial were determined by TEM and SEM, followed by DLS measurements and zeta potential examination.
The gold theranostic agent was tested in vitro against mouse and human PDAC cell lines (Panc1, Pan_O2, AsPc1). Metabolic activity and cell number were assessed after exposure to AuNRs-GEM and light in both hypoxic and normoxic conditions. A C57BL/6J mouse PDAC model was established utilizing the Pan_O2 cell line. Tumor oxygenation was measured using electron paramagnetic resonance (EPR) spectroscopy, employing Jiva-25 or Bruker E540L instruments, with trityl OX071 or Oxychip as the spin probe. Tumor anatomy and vascular structure were evaluated using ultrasound, including B-mode and Power Doppler with Vevo F2 from FujiFilm VisualSonics.
The therapeutic approach involved administering AuNRs-GEM (gold nanorods loaded with gemcitabine) combined with hyperthermia using near-infrared light (~808 nm). The treatment regimen included six doses of AuNRs (approximately 1 μg/ml) along with GEM (approximately 45 mg/kg BW) and near-infrared heating (approximately 3 x 1.5 min during 12 min) administered every 72 hours. All experiments obtained approval from the Local Ethics Committee (no. 151/2022 and no. 250/2020).

Impact

The developed compound holds the potential to enhance the efficacy of unresectable PDAC in the future. pO2 can be predictive of therapy results.

Acknowledgment

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Yes

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